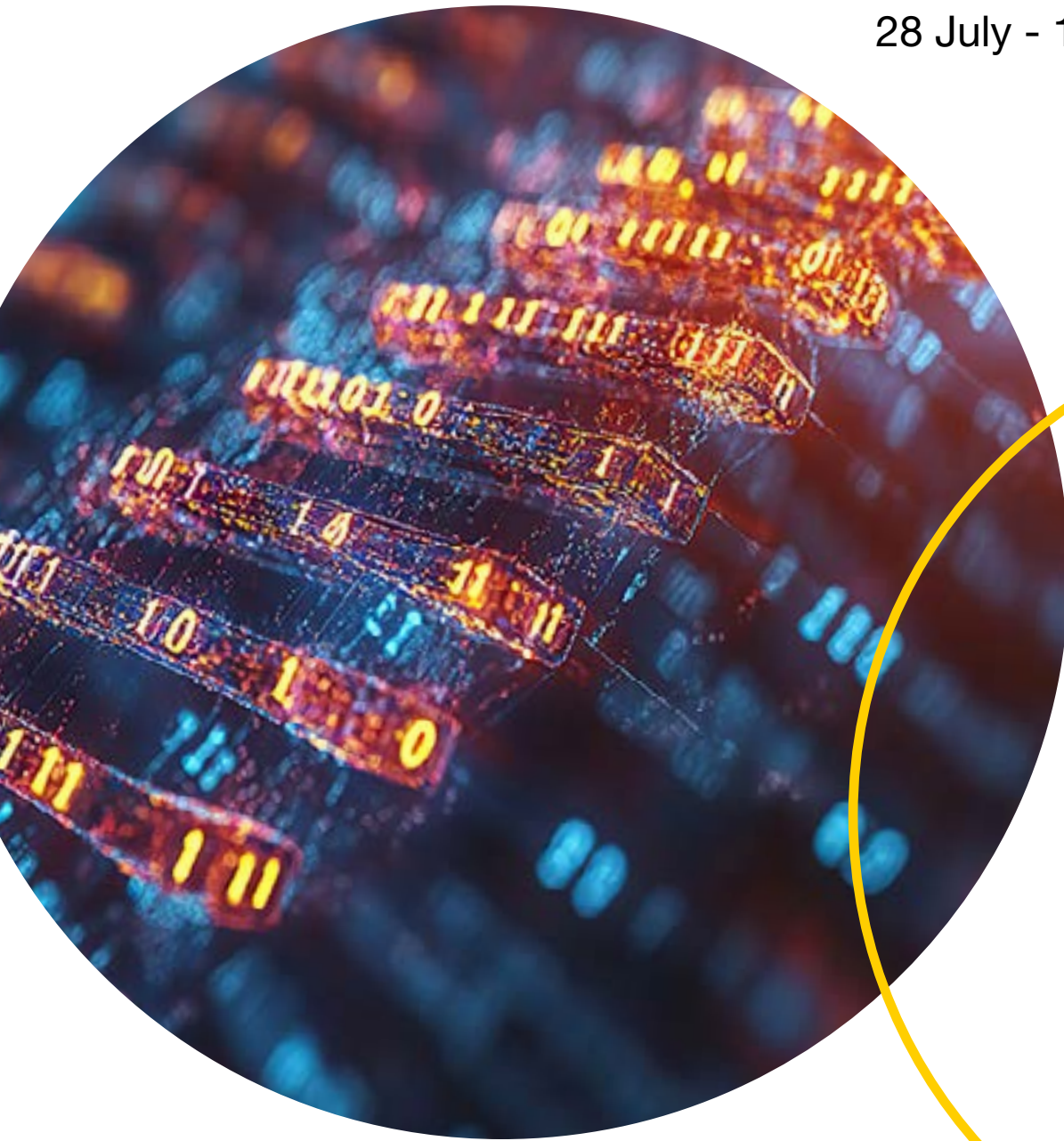


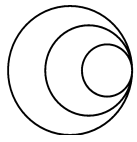
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# **CRISPR screening and analysis - virtual**

28 July - 1 August 2025



**course information**



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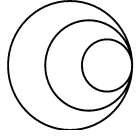
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# **CRISPR Screening and Analysis - virtual**

28 July - 1 August 2025

Virtual via Zoom



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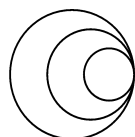


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# **Wellcome Connecting Science CRISPR Screening and Analysis - virtual**

**28 July - 1 August 2025**

**course information handbook**



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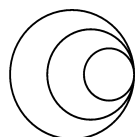
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## Staff

### Wellcome Connecting Science Learning and Training - Courses team

|                     |                                      |
|---------------------|--------------------------------------|
| Alice Matimba       | Head of Training and Global Capacity |
| Cassandra Soo       | Laboratory Courses Manager           |
| Martin Aslett       | Informatics Manager                  |
| Amy Pass            | Event Organiser                      |
| Christopher Adamson | Laboratory Operations Officer        |
| Aaron Dean          | Laboratory Technical Officer         |
| Vaishnavi Gangadhar | Informatics Technical Officer        |

URL: <https://coursesandconferences.wellcomeconnectingscience.org/our-events/courses/>  
Email: [courses@wellcomeconnectingscience.org](mailto:courses@wellcomeconnectingscience.org)

## Training team

### Organisers

|                     |                                             |
|---------------------|---------------------------------------------|
| Matthew Coelho      | Wellcome Sanger Institute, United Kingdom   |
| Kalpana Surendranth | University of Westminster, United Kingdom   |
| Panos Zalmas        | Open Targets Validation Lab, United Kingdom |

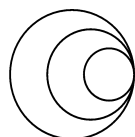
### Instructors and assistants

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| Khalid Akram          | University of Westminster, United Kingdom                              |
| Jamie Billington      | Wellcome Sanger Institute, United Kingdom                              |
| Saamin Cheema         | Wellcome Sanger Institute, United Kingdom                              |
| Cansu Dincer          | Wellcome Sanger Institute, United Kingdom                              |
| Saroor Patel          | Wellcome Sanger Institute, United Kingdom                              |
| Munuse Savash         | Nuffield Department of Women's and Reproductive Health, United Kingdom |
| Sandeep Sundara Rajan | Wellcome Sanger Institute, United Kingdom                              |

## Guest speakers

We are grateful to the following guest speakers for their time and contribution to this course:

|                             |                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------|
| <b>Andrew Bassett</b>       | Wellcome Connecting Science, United Kingdom                                           |
| <b>Manuel Kaulich</b>       | Institute of Biochemistry II - Goethe-Universität Frankfurt am Main, Germany          |
| <b>Eloise Lockyer</b>       | Open Targets Validation Lab, United Kingdom                                           |
| <b>Alice Yunsi Refermat</b> | Center for CRISPR Target Discovery -IGI Functional Genomics, United States of America |



## **CRISPR Screening and Analysis - virtual**

### **Programme**

Pooled and arrayed CRISPR screens have diverse clinical and research applications. Arrayed screening involves targeted gene editing for investigating specific gene function and biological pathways, whereas pooled screening enables a rapid high-throughput global approach to gene discovery.

This two-part course follows on from the 2024 CRISPR Cellular Generation and Modelling course and offers participants an opportunity to explore the techniques, nuances, and applications of CRISPR screening technologies.

The virtual component provides a comprehensive overview of the theory, design, and practical analytical skills relevant to CRISPR screens with examples of both pooled and arrayed screens and their applications.

It consists of hands-on computational practicals, lectures, and seminars given by international experts. Additionally, a group project will be assigned to participants for completion as part of the curriculum.

The material delivered will be focused on:

- The fundamentals of CRISPR screening
- Experiment and library design
- Applications of pooled and arrayed screens
- Quality control and validation of pooled and arrayed screens
- Screening informatics

### **Learning outcomes**

After attending this course, participants should be able to:

- Use a variety of online experimental design tools and resources for screening applications
- Explain the differences in methodology and application of pooled versus arrayed screens
- Design a screening project for a given gene and explain the strengths, limitations, and strategic decisions for the intended application

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